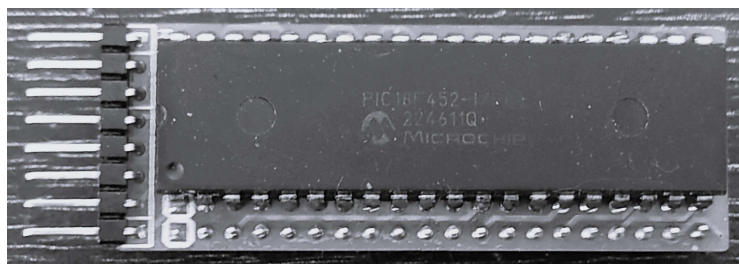


FT-757GX FTPLUS OPERATION SUPPLEMENT

**(c) 2026 Matthew Bostock M0WCA
Designed, manufactured and programmed in England**

v1.7.0

FTPLUS REPLACEMENT CPU SYSTEM FOR THE FT-757GX



GENERAL DESCRIPTION

This is a replacement microprocessor package for the Yaesu FT-757GX mobile transceiver. This has been created because the original CPU for this radio has been out of production for many years.

This has been built upon many months of work and testing and it continues to be actively developed.

Using VK2TRP's project as a start point, it has been slowly improving and evolving until it has reached its current level.

The package involves two parts. The firmware which this document is outlining – and the main PCB itself.

The part is based on a DIP-40 PIC IC. These are low cost, easy to program and easy to incorporate into a design like this. PIC18F452 (and PIC18F4520 from v1.4.5 onwards, without FTPlus bootloader) chips are supported. More can easily be added.

The firmware attempts to faithfully recreate the functions of the original CPU, while providing some useful functional upgrades.

Personal note from the author:

PLEASE: Don't replace working CPUs with this. These radios are getting old and value will be maintained if kept original. This was designed for the many radios that have died decades ago and are languishing on shelves facing an uncertain future. I understand that having new features is tempting, however I cannot imagine the FTPlus family of parts would increase a radios value.

That said, if you have fitted a part, regardless of reason, WELCOME!

Check out the other FTPlus parts available, should you need them: Our display CPU and VFO rotary encoder assemblies.

Cheers, God bless,

Matthew M0WCA

FTPLUS FEATURES

- 14 Memory channels
- 20 individual VFO band save points
- Enhanced CAT interface (Yaesu and Kenwood emulation, half-duplex)
- Multiple Baud rate options
- CB Mode with 8 worldwide bands and clarifier
- Accelerated fuzzy logic dial
- Fully manual “button tuning”
- Off-frequency correction
- Fine tuning mode
- PC application with bootloader for easier upgrades without a programmer
- Enhanced PMS scanning system
- Fully open-source
- Fully compatible with all display CPU chips

THIS MANUAL DOES NOT REPLACE THE ORIGINAL YAESU INSTRUCTION MANUALS.
IT IS INTENDED TO SUPPLEMENT THAT DOCUMENT.

WARNING

You undertake the modification of your equipment at YOUR OWN RISK. The author will NOT BE HELD LIABLE FOR ANY DAMAGE, HOWEVER CAUSED.

Ensure you transmit within bands you are licenced for. It is against the law in most territories to transmit using a non type-rated transceiver on CB bands.

The creator of FTPlus does not encourage or condone transmission on CB bands using high-powered equipment. This may lead to severe civil or criminal penalties, including the confiscation of equipment, fines or incarceration.

PTT OVER CAT WARNING

WHEN TESTING CAT REMOTE PTT FUNCTIONS, ENSURE A 100W DUMMY LOAD IS CONNECTED TO THE ANTENNA SOCKET, ALSO USE THE LINEAR SWITCH ON THE BACK OF THE RADIO TO DISABLE POWER OUTPUT WHEN TESTING. YOU HAVE BEEN WARNED.

IF CAT PTT MODIFICATION IS FITTED, WHEN UPDATING FIRMWARE WITH PICKIT 3, PRESS THE LINEAR BUTTON TO SWITCH OFF THE FINAL OUTPUTS.

UPGRADING THE FIRMWARE WITH THE CHIP FITTED IN THE RADIO, MAY ACTIVATE PTT IF YOU HAVE CAT PTT FITTED, PARTICULARLY IN OLDER FTPLUS BUILDS (v1.6.0 and under)

V1.6.0+ has certain safeguards built-in that “should” prevent any PTT activation when updating via Pickit / MCLR is held low.

RF transmission is only allowed when the PLL chips are properly tuned. By waiting until FTPlus has fully booted before we tune these PLL chips, RF transmission is prevented during Pickit upgrade.

Early versions of FTPlus tuned the PLLs immediately at boot, which caused RF transmission.

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INITIAL SETUP

CAT mode

When switching on your radio for the first time, the default baud rate is 4800.

This can be changed when you first switch on your radio or in the FTPlus configurator. This is the procedure:

Power off transceiver
Hold M>VFO button
Power on the transceiver

You will see a baud rate in the display (default is 4800)

Rotate dial, or use up/down buttons to change your desired baud rate.

To save, press M>VFO again and the radio will boot up.

If FTPlus detects the configurator, baud rate will stay at 4800. It is all automatic. No baud changes are required to connect to the configurator. It is automatic.

So you are free to set whatever baud rate you like without worrying about affecting a connection to the configurator app.

CAT mode changes are handled automatically. No need to set Yaesu or Kenwood mode! You just need to set your COM port, stop bits and (if not using configurator) baud rate.

Hint!

On a fresh boot or EEPROM reset, all memories are empty, no offsets are saved and everything is “stock”.

If you would like to use the configurator, its important to have the configurator open and listening for the “magic packet” BEFORE the radio is switched on.

Turning power off, then selecting COM port will make this easiest! Once we have done this, THEN we power on the radio.

If all goes to plan, FTPlus and the configurator app should connect automatically.

MEMORY PROGRAMMING

To save a tuned frequency to the onboard memory bank, we follow the same procedure as the original CPU.

In VFO mode (VFO A or B), tune to your desired frequency and press VFO>M.

The current memory channel location will flash and beep TWICE in the display.

In addition, you can save from a third VFO frequency, when MR mode is active.

Channels are numbered from 0 to E, so when you reach 9, any further increases will be in letters, so one after 9 is 'A'.

In CB mode, saving of CB frequencies directly to the memory bank isn't implemented, as one can just save these frequencies from VFO mode if desired, by pressing M>VFO in CB mode.

HINT!

In VFO mode:

You can change memory channel WITHOUT entering MR mode. Just hold BAND UP or BAND DOWN for 1.5 seconds to increase or decrease the current channel. There will be two flashes and beeps, showing the new channel.

In MR mode:

You can tune up and down the entire frequency range, without disturbing your VFO A or B contents, making a "third" VFO. Frequencies tuned using this feature, are NOT saved at all, unless the VFO>MR button is pressed.

You can also move up and down the bands, by pressing and holding BAND UP/DOWN for 1.5s.

A short press on BAND UP/DOWN moves the current memory up or down by one.

In CB mode:

To transfer a CB frequency to VFO/MR, press M>VFO when in CB mode. This will save the CB frequency to current VFO, allowing a save to MR if required.

To move a saved frequency to VFO A or B, press M>VFO to move, or M<>VFO to swap.

This works regardless of VFO or MR mode. When the chosen button is pressed, the display will flash and beep, before the frequencies are swapped or moved.

Ensure you have your desired MR and VFO selected before pressing this button, to avoid any mistakes.

VFO / MR / CB Transceiver Mode*

Three modes exist within FTPLUS. All modes are accessed by pressing MR/VFO.

To toggle between VFO and MR modes, a single short press on MR/VFO will change modes.

To enter CB Transceiver mode, hold down MR/VFO for approximately 3 seconds, in Wideband mode only.

VFO A/B – VFO tuning

When in this mode, press BAND UP / DOWN to move from one frequency band to the next. **Current frequency will be allocated to the current VFO memory. This effectively gives an extra 20 memory channels, overwriting the stock “bands”.**

To reset these to stock values, hold VFO>M for 2 seconds. Both VFOs are treated separately, using their own band save points and their own buffers.

Press and hold BAND UP / DOWN to increase or decrease memory channel (hidden in VFO mode of course) – this is handy for saving frequencies to memory without entering MR mode first.

MR – Memory Recall

Tune the whole frequency range. MR will be shown in the display, along with the current channel.

To change channel, a short press on BAND UP/ DOWN will change selected channel up or down.

Press and hold BAND UP/ DOWN to increase or decrease CHOSEN BAND. This is a convenience feature, allowing you to change bands in preparation for saving, using the extra VFO we use within MR mode.

When in MR mode, you can tune the radio as you wish, WITHOUT affecting VFO A or B.

HINT!

Saved VFOs can be quickly restored to stock values by holding VFO>M for 2 seconds. No other settings are changed or deleted.

CB Transceiver Mode* – Worldwide

Use VFO wheel, microphone or BAND UP / DOWN to change channels.

Channel number is displayed, together with a band number.

Band 1 – Super super LOW (25615 – 26055 Mhz)

Band 2 – Super LOW (26065 – 26505 Mhz)

Band 3 – LOW (26515 – 26955 Mhz)

Band 4 – MID (26965 – 27405 Mhz)

Band 5 – HIGH (27415 – 27855 Mhz)

Band 6 – Super HIGH (27865 – 28305 Mhz)

Band 7 – Super super HIGH (28315 – 28755 Mhz)

Band 8 – UK CB (27.60125 – 27.99125 Mhz)

To view your chosen frequency, press CLAR. This will switch on the CB clarifier.

In this mode, the frequency can be adjusted. Clarifier frequency is not saved automatically.

To change bands

Press SPLIT or Mic FAST to go up one band. If you reach the last band (8), band will return to lowest (1).

HINT!

To quickly scan all CB bands, press in 500k STEP switch before pressing PMS in CB mode.

To copy a CB frequency to VFO A or B, press M>VFO when in CB mode.

*** CB Mode is only available when radio is in WIDEBAND condition!**

VFO Button Tuning Mode

When in VFO mode, it is possible to manually tune to ANY FREQUENCY, in any increment, without using the VFO dial and without using a computer.

Frequencies are immediately tuned on-the-fly.

Entering Button Tuning Mode

1. Ensure transceiver is in VFO mode (VFO A or B).
2. Ensure DIAL LOCK (D LOCK) is enabled and LOCK is illuminated in the display.
3. Hold down D-LOCK for approximately 1.5 seconds. The transceiver will beep. Release D-LOCK.
4. The display will show the frequency, with the left-most digit flashing.

Operation

5. Press BAND UP / DOWN or MIC UP / DOWN to change value as required. Hold to go up and down fast.
6. Press D-LOCK or MIC FAST momentarily. The cursor / flashing digit will move to the right 1 place.
If you get to the 10hz digit, pressing D-LOCK will put the cursor back to 10 x MHZ (where we started).
Repeat step 5.
7. You will notice all the digits can be changed and altered as you like, the hidden digit will scroll into view.

Fast Tuning

You can hold down any BAND button or MIC button to count up or down quickly.

Exiting Button Tuning Mode

Press VFO A/B or hold MIC FAST to exit and save to current VFO

Example:

1 2 3 4 5 6
Flash

Press D-LOCK momentarily

1 2 3 4 5 6
Flash

Press D-LOCK momentarily

1 2 3 4 5 6
Flash

Press D-LOCK momentarily

1 2 3 4 5 6
Flash

Press D-LOCK momentarily

1 2 3 4 5 6
Flash

Press D-LOCK momentarily

1 2 3 4 5 6
Flash

Press D-LOCK momentarily (Last number visible)

2 3 4 5 6 7
Flash

Press D-LOCK momentarily (Loop back to start)

1 2 3 4 5 6
Flash

Hint!

CAT mode can be used in button tuning mode. If you make a frequency change in your program, it will be reflected in Button Tuning mode.

Microphone Button Functions

Microphone button action has been improved in FTPLUS, compared to the standard Yaesu settings.

The function of these buttons is described below.

In virtually all functions that require up and down, you can use the microphone.

VFO Mode

UP/DOWN (Single Press)

In fine-tune mode, frequency is changed by 10hz.

UP/DOWN (Holding)

Frequency is changed by 100hz.

FAST + UP/DOWN

Frequency is changed by 1Khz.

FAST (Single Press)

Band increase. (Looping).

FAST (Holding)

Activate manual button tuning mode.

MR Mode

UP/DOWN (Single Press)

Channel Up/Down by 1.

UP/DOWN (Holding)

Channel Up/Down quickly.

FAST (Holding)

Activate manual button tuning mode (will return to VFO mode upon exit)

CB Mode

UP/DOWN (Single Press)

Channel UP/DOWN by 1.

UP/DOWN (Holding)

Channel UP/DOWN quickly.

FAST + UP/DOWN

No Function

FAST (Single Press)

Change Band(FCC / CEPT - BRITISH)

FAST (Holding)

Exit CB Mode

HINT!

Serial CAT transmit is shared with MIC DOWN signal. So if you are using bidirectional Kenwood CAT and you press MIC DOWN button, this will cause a zero to be sent to the data stream.

This will not cause any problems or faults to the radio, as the data is self-correcting once the DOWN button is released.

All data packets are a certain length and contain validation / termination characters, so if any are invalid, they get rejected.

TUNING DIAL TYPE SELECTION

Within FTPLUS, there are two VFO dial MODES

To change dial type, ensure you are in VFO or MR mode and HOLD M>VFO for approximately 3 seconds.

DIAL A – STANDARD DIAL

This is the standard Yaesu-specification dial. One revolution is equal to approximately 10khz.

DIAL B – ACCELERATED DIAL

This dial will move up and down faster, the more “effort” is put in to spinning the tuning wheel. Spinning quickly will increase the frequency counting speed faster or slower.

Speeds are handled seamlessly and automatically.

FINE TUNING MODE

In VFO or MR mode, fine tuning can be activated.

To activate or deactivate Fine Tuning, hold down D-LOCK until a beep is heard

When this mode is active, small turns on the VFO tuning wheel will make the display move one digit to the LEFT, allowing the missing digit to be displayed

For example, when tuned to 7.000.0 mhz, turning the dial will make the frequency display move one digit left while tuning, so the display would display 70.000.1, 70.000.2 etc. The display will hide the extra digit when tuning stops.

This gives approximately 5khz increase / decrease in frequency per revolution.

This mode also works when tuning using the MIC buttons.

FREQUENCY CORRECTION AND ALIGNMENT

FTPLUS has an alignment feature, which allows you to set, for example a ZERO POINT if your radio is not quite aligned correctly

Example:

You tune to 27.555 and all stations sound high-pitched. This is because your radio is slightly off-frequency.

The procedure is outlined here:

Tune to a desired frequency, using fine-tune if necessary to get to true ZERO.

Activate the alignment mode and slowly turn the tuning dial clockwise or anticlockwise, until the stations sound correct. The difference between your displayed frequency and the true frequency is called an OFFSET.

An offset can either be +/-.

The number doesn't really matter, but when the stations are sounding correct, you press the SAVE button to store the offset. This will then be applied wherever you tune.

There are even two modes for this.

Default mode is one single offset for the whole tuning range.

You can also activate "per-band" offsets, so you can set different offsets per band.

Handy if you need one offset for 40 meters LSB and a different one for 20 meters USB!

You can even enable RX, TX or both (default) frequencies to apply an offset to. This is accessed by CAT (Y-Library).

If your radio is out of tune by approximately the same amount across the whole tuning range, single offset mode is sufficient

If it varies from band to band, activate multiple offset mode.

To toggle offset modes, hold down VFO<>M. You will hear either two or three beeps.

Two beeps is SINGLE OFFSET MODE

Three beeps is PER-BAND OFFSET MODE

2) TUNE TO A KNOWN ON-FREQUENCY SIGNAL. IGNORE HOW IT SOUNDS OFF FREQUENCY! TUNE TO WHAT WOULD BE ITS CORRECT FREQUENCY.

Ensure you get it as close as possible to 0 in the hz digit of the display. It will sound incorrect, or off-frequency as we have not yet aligned it.

Activating Fine Tune can help and make things even more accurate. Being on ZERO, means you have found ZERO POINT. The display is on Zero point, but the radio tuning circuits are not. We correct this now.

HOLD DOWN CLAR for approximately 3 seconds.

The display will show 000.

With the station broadcasting in the background, adjust the tuning dial until the most "correct" sound is heard.

When happy, hold down CLAR for 3 seconds until you hear a beep. The tuned frequency should now match your displayed frequency.

HINT!

To reset to zero, press D-Lock when in alignment mode.

For a reminder of your chosen display frequency, press CLAR once quickly.

CAT INTERFACE AND MODES

FTPLUS supports two methods of CAT communication.

1) Legacy Yaesu FT-757GX mode

<u>RS232:</u>	<u>Receive from PC only</u>
<u>Baud Rate:</u>	<u>4800 Baud</u>
<u>Stop Bits:</u>	<u>2</u>

No other cables are required. Use the original CAT port on the back of the FT-757GX

Some extra commands have been added, which could be turned into a custom command in FLRIG

See Page 14.

2) Advanced Kenwood CAT emulation

<u>RS232:</u>	<u>Bidirectional</u>
<u>Baud Rate:</u>	<u>1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200</u>
<u>Stop Bits:</u>	<u>2</u>

FTPlus emulates a Kenwood TS-140 in Kenwood CAT mode. This requires a “serial TX” transmit line to your RS232 USB device.

On FTPLUS boards, this extra TX connection is provided on the circuit board itself.

This transmit line is provided on the PCB of FTPLUS. labelled RS232 TX. This is also connected to MIC DOWN button on the front panel, (PIN 3 MIC socket). It is for the operator to decide the best way of attaching the TX line.

Hint!

Kenwood mode ONLY:

The **Y Library** gives CAT access to ALL front panel controls and FTPlus functions. These commands could be added to FLRig etc... Details on Page 27.

Connection Examples:

Use the CAT port on the back of transceiver as normal and piggyback a wire to MIC DOWN on the front panel, for serial transmit. You could even attach this to your hand microphone plug.

Or

Feed 3 x DuPont or similar cables to the RS232 port on FTPLUS and fit these between the casing of the radio, for RX, GND and TX serial.

Hint!

CAT mode can be completely paused, yet remain connected to computer by holding SPLIT for 3 seconds. To resume CAT communication, simply hold SPLIT again.

Note the limitations of Yaesu mode CAT. The radio can only receive commands. It cannot send them. This is the original design of the FT757GX.

CAT INTERFACE AND MODES

LEGACY YAESU MODE

Hint!

If you wish to use legacy Yaesu CAT mode and your radio is in Kenwood mode, close all apps using Kenwood CAT and turn radio off and on, or send 'YA;' via serial.

Please see the original FT-757GX manual for information regarding CAT commands and formatting.

Some NEW COMMANDS have also been added for your convenience.

F1 = 100 & 10 Hz

F2 = 10Khz & 1 Khz

F3 = 100Khz & 1Mhz

F4 = 10Mhz & 100Mhz (100Mhz = not used)

CM = Command

XX = Value not important, is ignored

Command: E0, E1, E2, E3, E4, E5, E6, E7, E8, E9, EA, EB, EC, ED, EE

Requires (F1 F2 F3 F4 E0) – (F1 F2 F3 F4 EE)

This is for fast memory saving. E0 – EE are the channel numbers (0 – 14).

Command: 0A (for reference)

Requires (F1 F2 F3 F4 0A)

Setting of frequency for current VFO

Command: 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 1A, 1B, 1C, 1D, 1E

Requires (F4 F3 F2 F1 10) – (F4 F3 F2 F1 1E)

Fast memory saving, in human-readable format. 10 – 1E are the channel numbers (0 – 14).

EG 7.000mhz to channel 3 = 0700000013

EG 10.100Mhz to channel 10 = 101000001A

Command: 0F (for reference)

Requires (F4 F3 F2 F1 0F)

Setting of frequency for current VFO in human-readable format

OTHER COMMANDS

Command: FC

Requires (XX XX XX XX FC)

Temporary wideband, until next reboot.

If activated when transceiver is widebanded via the internal switch, will emulate non-widebanded condition until reboot. This is useful for testing perhaps.

Command: FD

Requires (XX XX XX XX FD)

CB Mode on/off (only in widebanded mode, either temporary or permanent wideband)

Command: FE

Requires (XX XX XX XX FE)

Full EEPROM reset

ADVANCED KENWOOD CAT EMULATION

Ensure the serial transmit TX pin is connected as outlined in the previous section

Kenwood commands are sent as ASCII, with a ';' as a terminator

The correct CAT mode is automatically detected.

Kenwood CAT emulation relies on ';' (0x3B) as a terminator, so if anything is received by the radio CAT interface containing ';', Kenwood mode will be activated.

If you are in Kenwood mode and you wish to use legacy Yaesu CAT mode, close all apps using Kenwood CAT and turn radio off and on, or send 'YA;' via serial.

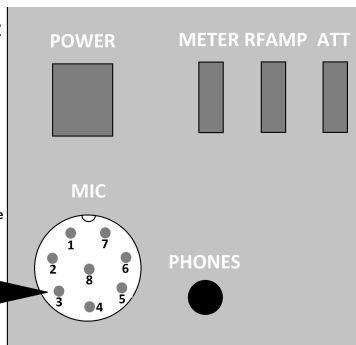
Most Kenwood TS140 commands are supported, with more added as development progresses.

CAT RS232 Transceiver to PC connection (TX)

No modifications or external wires necessary. MIC DOWN connection. MIC DOWN functionality still working!

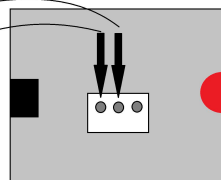
Use the original CAT socket on the back of transceiver for RX and GND (See Yaesu manual). Connect to MIC socket. Even connect inside microphone / cable!

**CAT RS232 TX
(To PC / USB
serial)**



**CAT RX
GND**

Yaesu manual says CAT RX goes to a buffer. This is incorrect. It is connected directly to CPU!



Hint

Read the Y Library Glossary on Page 27 for details about FTPlus commands.

The Y Library is in addition to regular Kenwood commands

You can send the command 'IF;' and if working, you should get a reply starting IF.....;

Hardware DIAL LOCK is disabled while CAT PTT is transmitting, changing to a software dial lock.

Hardware dial lock will be re-enabled after CAT PTT ends, if D LOCK is switched on.

Y Library is a library of (nearly?) all commands in use within FTPlus.

If 'Soft PMS' or 'Soft 500k' in Y-Library are active, pressing the actual physical PMS or 500k STEP button will override this setting.

Offset alignment is controlled by sending the offset straight to FTPlus (YFxxxxx; command).

The only function that cannot be activated over CAT is Manual Button Tuning, as there is no use case to access this feature over CAT and is considered a waste.

NEW PMS SYSTEM

The PMS system is a scanning system, designed to scan through frequencies quickly and stop when a signal is received.

There are two PMS controls.

The PMS switch itself.
The 500K button.

The PMS switch turns the feature on or off
The 500K button changes how the PMS scan is performed.

The different PMS modes are outlined on this page.

VFO MODE – Press PMS to begin and to end scanning

500k button pushed OUT:

Scanning will begin from the current frequency, to the start of the NEXT band.

Example: You are tuned to 7.165Mhz.
Scanning will begin from 7.165 mhz and carry on until the start of the NEXT VFO, which is 10,100mhz, before looping back to start of current band, which is 7.000Mhz.

500k button pushed IN:

Scanning will begin from the start of the current CEPT/FCC band, until the end of the band.

Example: You are tuned to 7.165mhz.
Scanning will begin from 7.000mhz and carry on until the end of the 40 meter band, which is 7.200 mhz

Most, if not all CEPT/FCC bands are programmed. Just tune to anywhere within your chosen band and PMS will do the rest

MR MODE – Press PMS to begin and end scanning

500k button pushed OUT:

Scanning will begin from frequency stored at current memory channel, to the frequency stored at the NEXT channel.

Example:

You have 14.150 stored to channel 2

You have 7.165mhz stored to channel 3

You have 10.150mhz stored to channel 4.

If you are currently tuned to channel 3, pressing PMS will scan from between channel 3 to channel 4. So the scanning range will be from 7.165 to 10.150, going UP. Once 10.150 is reached, the cycle begins again from 7.165.

If you are currently tuned to channel 2, pressing PMS will scan from between channel 2 and channel 3. So the scanning range will be from 14.150 to 7.165mhz, going DOWN. Once 7.165 is reached, the cycle will restart from channel 2 (14.150).

If the SAME FREQUENCY is stored, eg 7.000 is stored as default in channel 0 and channel 1, you will hear an error beep because scanning isn't possible, as the start and end frequencies are the same.

500k button pushed IN: ***

All channels are scanned individually, beginning at Channel 0 and ending at Channel E, then starting again.

***** This can cause rapid relay clicking and it isn't recommended to use**

NEW PMS SYSTEM contd.

CB MODE – Press PMS to begin and end scanning

500K STEP button out / not enabled

PMS will begin scanning from current channel until the last channel of the current band, then from channel 1 of current band.

500K STEP button in / enabled

PMS will begin scanning from current channel until the last channel of the last band, then from channel 1 of first band.

See CB mode section for band information.

To scan a different band, ensure PMS is stopped. Change transceiver to desired band and press PMS to start scanning.

HINT!

Current frequency or channel can be skipped, by pressing BAND UP / DOWN or MIC UP/ DOWN.

If PMS is stopped while frequencies are changing (ie squelch closed), the previous frequency BEFORE PMS will be retuned.

If PMS is stopped while listening (ie squelch open), the current frequency will remain tuned upon exit from PMS mode.

OTHER FUNCTIONS

EEPROM RESET

To reset the EEPROM back to defaults, with the radio switched on, press and hold VFO A/B until beeps are heard.

Please turn the radio OFF then ON again.

BAND RESET

If jumbled up “bands” are causing confusion, hold VFO>M when in VFO mode, to set to default.

No other settings are altered. This only resets the “VFOs” back to their stock positions.

AUTO FREQUENCY SAVING

By default, the current frequency is written to EEPROM (periodically, when frequency change detected, or when a change of memory mode is detected). This makes the internal battery within the radio obsolete.

This can be switched on and off if required by holding VFO>M when turning on radio. Frequencies will then ONLY be saved when a mode change is detected.

WIDEBAND INFORMATION

The radio can be widebanded using the switch behind the front panel, as usual.

If out-of-band transmit is attempted when transceiver is NOT widebanded, TX will be prevented until transceiver is retuned to a valid frequency. CB mode is also disabled.

TEMPORARY WIDEBAND

There is a temporary wideband facility, which is a command sent over Yaesu mode serial. See CAT section for more information.

This wideband will only persist until next power cycle reboot. See CAT section.

BUTTON ACTION QUICK REFERENCE

BUTTON	BUTTON PRESSED	BUTTON HELD
SPLIT	(VFO/MR MODE) SPLIT (CB MODE) BAND UP (1-8)	CAT PAUSE / UNPAUSE TOGGLE
VFO A/B	VFO A/B TOGGLE	RESET ALL (MEMORY ERASE!) TURN POWER OFF AND ON
MR/VFO	MR/VFO MODE	CB / EXIT CB
VFO<>M	VFO/MR SWAP QUICK EXIT CB (IF CB MODE)	*SINGLE ALIGNMENT ** BANDS ALIGNMENT
VFO>M	VFO/M SAVE	RESET VFOS (BANDS)
M>VFO	MR TO VFO A/B	*STANDARD DIAL **ACCELERATED DIAL
UP	UP BAND/CHANNEL	(IN VFO MODE) UP MEM CH (IN MR MODE) UP BAND
DOWN	DOWN BAND/CHANNEL	(IN VFO MODE) DOWN MEM CH (IN MR MODE) DOWN BAND
D-LOCK	DIAL LOCK	(LOCKED) – BUTTON TUNE (UNLOCKED) – FINE TUNE
CLAR	CLARIFIER	ALIGNMENT MODE / SAVE

HOLD SPLIT ON POWER-ON	CAT INITIAL SETUP
HOLD MR/VFO ON POWER ON	DISPLAY CAT SETTINGS (HOLD DOWN!)
HOLD VFO/M	ENABLE/DISABLE AUTO FREQ SAVING

CAT SPECIAL COMMAND GLOSSARY – SEE CAT SECTION.

YAESU MODE: NOTE: - - IS A DUMMY VALUE. ENTER ANYTHING)

XX XX XX XX [10 – 1E] (Ex. 02 77 81 25 1E = 27781.25 Ch E)	SIMPLE FREQ SAVE
XX XX XX XX [0F] (Ex. 02 77 81 25 0F = 27781.25 TO VFO)	SIMPLE FREQ SET VFO
- - - - - [FC] (Ex. 00 00 00 00 FC)	TEMPORARY WIDEBAND
- - - - - [FD] (Ex. 00 00 00 00 FD)	CB MODE TOGGLE
- - - - - [FE] (Ex. 00 00 00 00 FE)	EEPROM RESET

KENWOOD MODE

IE0; IE1;	CAT ID disable / enable change
Y Library;	Comprehensive command set

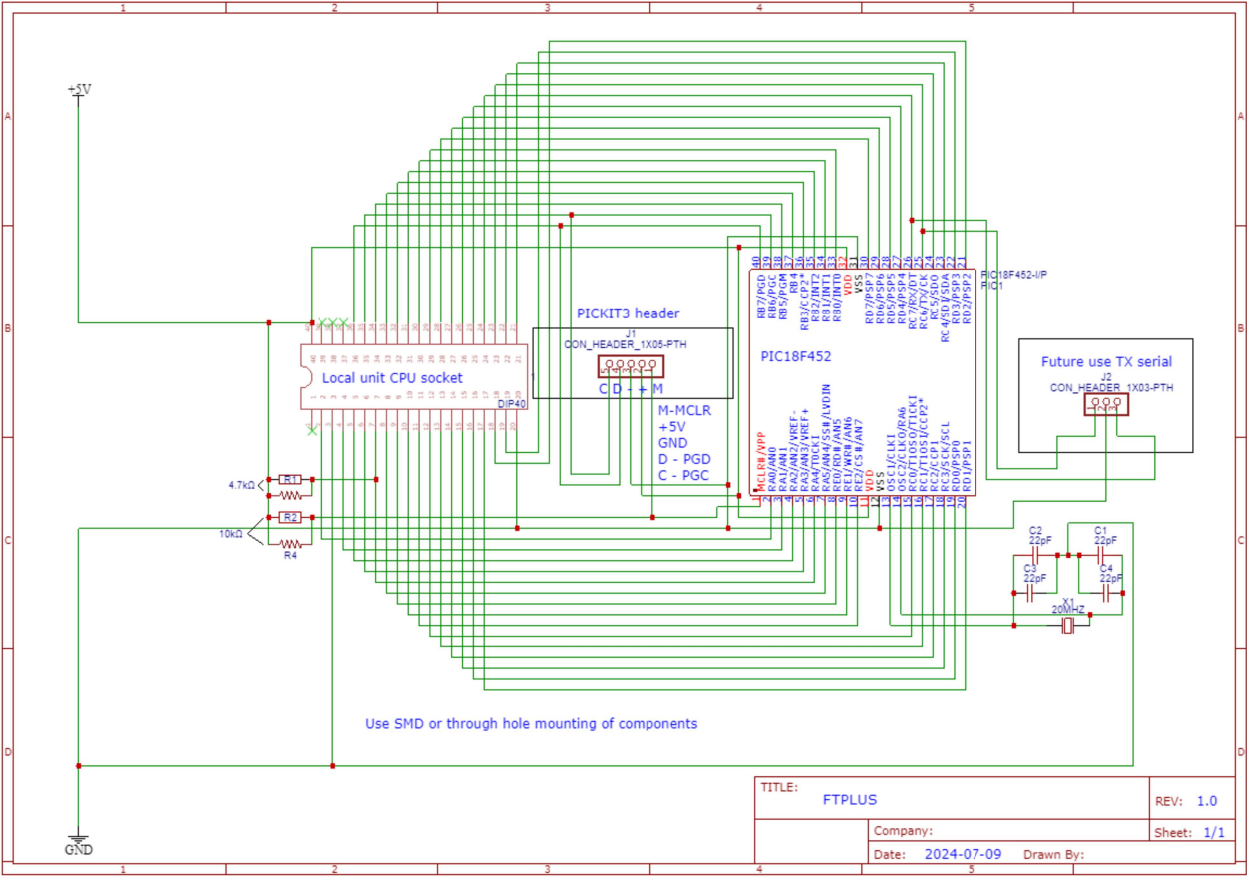
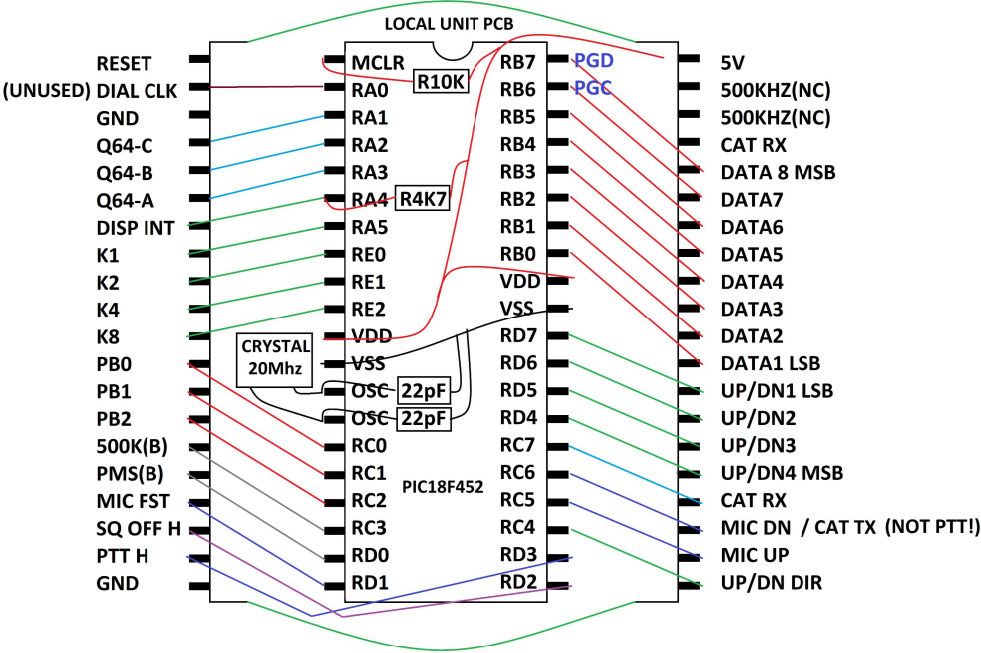
* 1 X BEEPS ** 2 X BEEPS *** 3 X BEEPS

SCHEMATICS, TECHNICAL INFORMATION, PARTS LIST

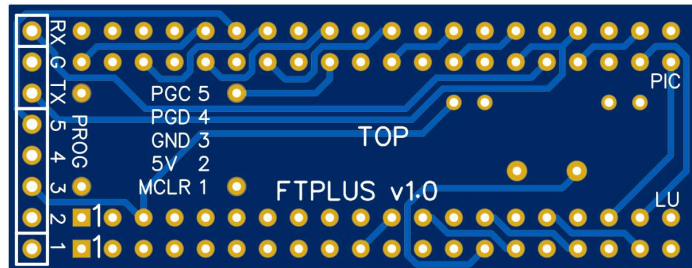
CONNECTION DIAGRAM AND SCHEMATIC

USE EITHER 0603 SMD COMPONENTS OR THROUGH HOLE. THE BELOW SCHEMATIC HAS 8 PASSIVE COMPONENTS. ONLY 4 ARE ACTUALLY USED.

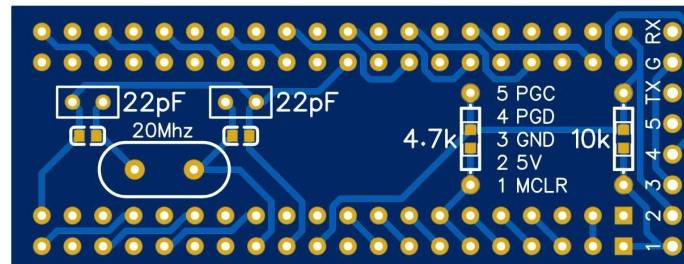
THIS IS TO SHOW SMD AND THROUGH-HOLE. YOUR CHOICE!



SCHEMATICS, TECHNICAL INFORMATION, PARTS LIST



UPPER VIEW



LOWER VIEW

***DISCLAIMER:** Board versions may update, but should all follow the same component parts lists and pinouts. Double-check your pinouts! A DIY FTPlus can also be built on prototype board, for an extra challenge. Follow the schematics on previous page.

Parts list: (Pads to accommodate THROUGH HOLE OR SMD components are provided.)

2x	22pF Capacitor – SMD 0603 / THT
1x	4.7K Ohm Resistor – SMD 0603 / THT
1x	10K Ohm Resistor – SMD 0603 / THT
1x	20 Mhz Crystal Oscillator
4x	20 Pin Turned Pin

Serial Pinout: (OPTIONAL. Recommended: Use CAT socket and MIC DOWN.)

RX	RS232 RX (Tied with CAT port on back of transceiver)
G	RS232 GND
TX	RS232 TX (Tied with Mic Down button on MIC connector - PIN 3)

PICKIT 3 Programmer Pinout: (If CAT PTT is fitted, read WARNING!!!)

5	PGC (RB6)
4	PGD (RB7)
3	GND
2	5V
1	MCLR

It is recommended that turned pins are used on the local unit and on FTPLUS.

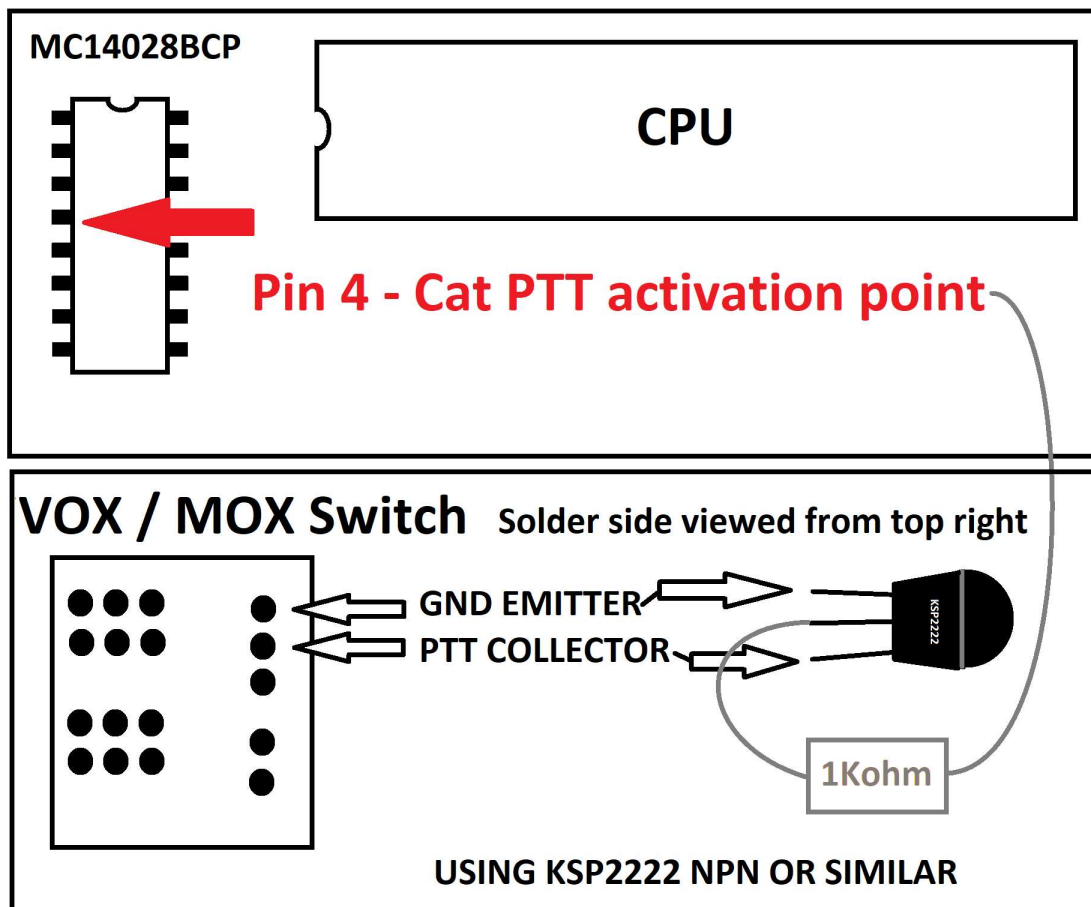
CAT PTT Modification – REFER TO WARNING ON PAGE 4

Parts required:

- Approx 4 inches (100mm) of 26 AWG wire, eg DuPont / IDC etc
- NPN transistor, at least 100mA (for example KSP2222, BC547 etc)
- 1000 Ohm resistor (1K)

Instructions:

- Radio turned OFF
- Remove radio top cover
- Remove FTPLUS for easier access if needed
- Locate IC marked MC14028BCP to the left of the processor
- Locate Pin 4 (going from top left). Solder length of wire to Pin 4.
- Solder 1K resistor to other end of wire
- Solder BASE of transistor to other end of resistor
- Locate MOX / VOX switch board (in front panel, directly under the left hand screw hole for the top lid), attach emitter and collector as shown



FIRMWARE UPDATE + BOOTLOADER

FT PLUS uses standard Pickit 3 pinout to reprogram. The programming pins are on the side of the PCB. These may be marked:

- **M** **MCLR** **1**
- **+** **5V** **2**
- **-** **GND** **3**
- **D** **PGD** **4**
- **C** **PGC** **5**

BOOTLOADER WITH UPDATER APP (Windows and Linux with Wine)

To make future upgrades easier, you can flash the official FTPLUS BOOTLOADER with a PICKIT 3 or similar device. Once you have done this, connect the CAT / RS232 TX and RX, then you can update over serial, using the **FTPlus updater tool**.

To install bootloader:

- Flash the correct FTPLUS BOOTLOADER HEX file using Pickit 3. Do not flash a BLHEX file, or serial will no longer work until you flash the correct bootloader. Most FTPlus boards will be using the PIC18F452 chip, however some may also use equivalent chips, eg PIC18F4520

To activate bootloader mode:

- Power off the radio
- Hold down PMS button and while holding it, turn on radio. Depending on radio, you may see the display filled with 00.000.0 – this is normal and a sign that the bootloader is running.
- **Run the FTPlus serial updater app and follow the instructions within that. Instructions are available by clicking Help in the app.**
- **If your PMS button is broken, you can activate bootloader mode by shorting pin 19 on the PIC chip to 5V source (pin 2 [second from bottom] on the FTPlus header pins maybe) and turning on the radio. Ensure you short the correct pin on the PIC CHIP ONLY! Remove the short when bootloader mode has been entered successfully.**

CAUTION:

- **FTPLUS v1.3 Onwards are supported. Do not attempt to flash earlier versions with serial, as bootloader will be overwritten, needing Pickit 3 to fix.**
- **CAT PTT is NOT REQUIRED to benefit from the bootloader. Only TX, GND and RX are needed, plus your USB-serial device.**
- **Ensure any file you flash is a BLHEX file, or has BOOTLOADER_ONLY in the filename. Failure to follow this WILL destroy the bootloader, requiring a Pickit to fix.**
- **Make sure you pick the version of FTPlus that matches your chip. You cannot do any harm by flashing the wrong one, it just will not work. If you do this, activate bootloader mode again and reflash the correct one.**

FIRMWARE UPDATE + BOOTLOADER

Basic FTPlus Updater Tool information:

If you have CAT mode connected (Kenwood mode), you can use the app straight away. **Clicking Help will give instructions on how to use it.**

Do NOT flash normal HEX files with the tool, unless you KNOW it is bootloader-safe. They will overwrite the bootloader. That's why we have BLHEX files!

This is the basic procedure:

Access bootloader mode, by **turning radio off, holding PMS, turn radio on, release PMS**
Choose your COM port and desired Baud rate. This needs to match in the app and the radio. There are two Baud speeds available. In bootloader mode, Baud speed is chosen by the 500k STEP button:

Button IN = 115200 Baud Button OUT = 4800 Baud

If 500k STEP is PUSHED IN, choose 115200 Baud in the app

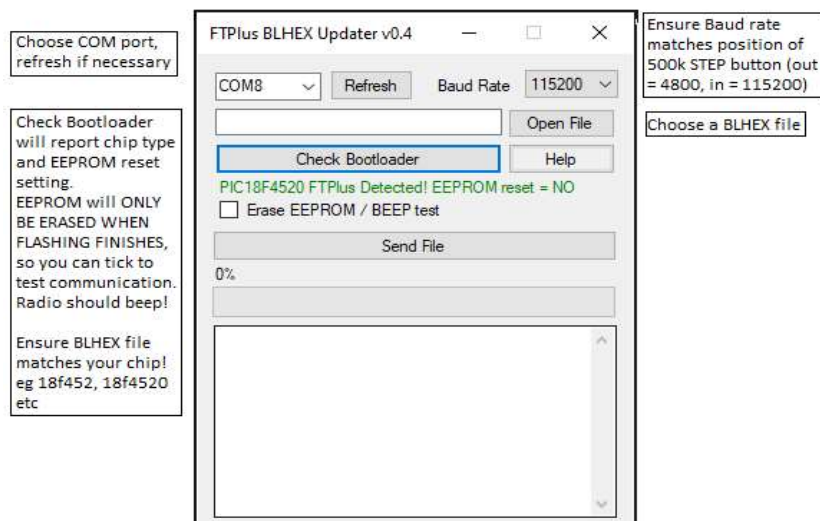
If 500k STEP is PUSHED OUT, choose 4800 Baud in the app

Check the state of the bootloader by clicking Check Bootloader – it should be green or blue as below

Choose your BLHEX file (this is the actual FTPlus firmware) and click Send File

To exit out of bootloader mode, turn radio off and on. Remember though a failed flash can look like you are in Bootloader mode (00.000.0 in display)... If you experience this, power off, hold down PMS to enter bootloader mode again and flash again.

Upon a successful flash, FTPlus starts running straightaway, automatically. There is no need to manually reboot the radio out of bootloader in most cases, apart from a failed flash.



Bootloader Status:

Unknown state. Is Baud correct? - Check Baud rate matches 500k STEP switch position

No reply. Restart bootloader or check COM port - No reply from bootloader. COM port incorrect, wiring to USB-TTL converter incorrect. Check all and restart bootloader by switching off, holding PMS and switching on

PICXXXXXX FTPlus Detected! EEPROM reset = NO - All is OK. EEPROM will NOT be reset

PICXXXXXX FTPlus Detected! EEPROM reset = YES - All is OK. EEPROM will be reset (recommended)

FTPLUS CONFIGURATOR APPLICATION

Introduction

The FTPlus Configurator application has been designed so that FTPlus settings can be changed as one wishes. You can also program memory channels and experiment with different filters and other items.

Scope and limitations

- This is designed as a helper app for those who wish to experiment. This is NOT a replacement for any CAT app you may already use.
- CAT TX and RX is required to use the configurator.
- The configurator requires sole access to the serial port. If running other serial-based apps, this will not work as the port is already in use.
- The following features are not available when the configurator is running:
 - PMS scanning
 - Offset adjustment / setting
 - CAT mute/pause
 - VFO-less manual button tuning

How does it work?

The configurator works by downloading the entire FTPlus EEPROM (256 bytes), applying offsets of the EEPROM to the app interface and allowing us to modify these, before uploading back to the FTPlus unit in the radio. The internal CAT commands to do this are as follows:
Heartbeat:

Running the configurator

- To run the configurator, just double-click and .exe file (Linux should be supported under Wine).
- Before running the app, power off the radio.
- Start the app and select your required COM port.
- You should see “Waiting for power-on magic”. This means the app is looking for a special handshake when we turn the radio on. Every time we turn our radio on, a “magic packet” is sent, at 4800 baud, in case the configurator is listening.
- If the configurator is listening, the CAT speed will be locked to 4800 baud and a EEPROM dump will be requested.

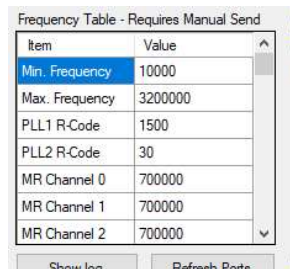
If the configurator sees FTPlus, as previously said, baud will lock to 4800(regardless of any custom baud rate) and there will be a flashing icon in the display and the words “FTPlus found” will be shown.

A heartbeat keep-alive signal is sent approximately every second from the configurator to FTPlus and FTPlus replies. If no reply is found after 3x retries, the radio reboots into “standard” mode and any custom baud rates are applied.

FTPLUS CONFIGURATOR APPLICATION

Frequency Table Items:

If editing frequencies, offsets, or anything else in the table below, a “manual” send is required.



Item	Value
Min. Frequency	10000
Max. Frequency	3200000
PLL1 R-Code	1500
PLL2 R-Code	30
MR Channel 0	700000
MR Channel 1	700000
MR Channel 2	700000

When editing is finished in the table, click “Manual send settings / frequencies” and everything will be uploaded.

- Min Frequency – Minimum accessible frequency in use by the radio. In Hz *10, so e.g 10000 is actually 100,000 Hz, or 100Khz.
- Max Frequency – Maximum accessible frequency in use by the radio. In Hz *10, so e.g 3200000 is actually 32,000000 Hz, or 32 Mhz.
- PLL1/2 R-code – These are reference codes for the onboard MC145145 PLL chips. More information can be found in the datasheet for these chips. Adjusting these “may” unlock other frequencies the radio can use. This is for experimentation only. Frequencies WILL alter and software realignment will be necessary. This is for advanced users ONLY. R-codes can be easily reset to default, so experimentation without opening the radio is quite safe(into a dummy load ;)...)
- MR Channel (0 – 14) – Set these to anything you like. No decimals, in Hz * 10. So if you wanted to program 27,555 Mhz, you would enter 2755500. If you just entered 27555, the frequency saved would be 27555 * 10Hz, which would be 275Khz.
- Single Offset / Offset <freq1 – freq2> – This is entered in *10Hz, up to a maximum of 99990Hz, preceeded with a + or -.
 - If in Per-band offset mode, single offset will not have any effect. Likewise if in single offset mode, per-band offsets will have no effect.
 - In per-band offset mode, only frequencies between the mentioned section will be affected by the offset. So if you applied an offset to 7.49Mhz, the offset wouldn't be active when going to 7.50Mhz as it is a different “band offset”.
 - So if you wanted to apply a positive offset of 100(1Khz) HIGHER than displayed frequency, you would enter +100. If you wanted 100 LOWER, you would enter -100.
 - This requires experimentation and maybe persistence!
 - It is recommended that offsets are applied via the Frequency Correction and alignment procedure on Page 12.
 - By default, all offsets are set at 0.

FTPLUS CONFIGURATOR APPLICATION

Table Items (contd):

- VFO A and VFO B bands 0 – 10. These are the bands that get altered as one tunes. We press the UP and DOWN button to move from band to band.
- When using FTPlus for a while, every user will have different numbers stored for bands, as the last frequency “tuned” is saved. You will notice that as you tune and select different bands, these numbers change.
- A “reset to default” will put these VFO bands back to stock, the same as holding VFO/M.

Once table items are modified, a “manual send” is required. These are NOT SENT automatically.

Other items:

Other settings are as follows:

- Normal Tuning / Accelerated tuning – selects standard or “fuzzy logic” dial (Page 11)
- Coarse Tune / Fine Tune – activates fine-tuning (Page 11)
- Single Offset / Per-band Offsets (Page 12)
- Periodic Saving / No Auto Saving – Enables periodic frequency saving when dial stops rotating (after approx 2 seconds)
- VFO A / VFO B / MR / CB – Selects memory mode (Page 8 / 9)
- BPF Filter Selection – This is for experimental purposes. There exists 4 (assigned to 10 values) band-pass filters. One or a combination of these are selected for a specific tuning range. This allows these filters to be changed for a particular tuning range. This may allow transmitting and receiving out of normal specification, depending on each radio.

Example:

- Filter 1 is designed for lowest tuning range, thus 0 – 1.5 Mhz. Filter 9 is the highest used for 25Mhz+... Changing “could” allow listening on other frequencies, in combination with different R-codes and tuning range modification (see earlier).
- This functionality is NOT GUARANTEED in any way. It will be different from radio to radio, even down to quality of components
- Also, each enhanced “long-press” FTPlus function can be assigned to a different button through drop-down menus.

Potential Issues

There may be times that the “magic packet” may not be received, or EEPROM can not be read.

In most cases, this is fixed by closing and restarting the app and/or powering the radio on and off.

You “may” lose all saved memories if not careful. A backup / restore settings feature is being planned.

The FTPlus configurator will improve over time. It is an assistance tool and is in NO WAY REQUIRED to fully use FTPlus.

Y Library Glossary

Cmd	Query	Set	Returns: Yxxxxxxx;	Function
YA	YA;	YA0;	YA0000001; - confirmation	0 - Yaesu mode (after sending command, will not accept Kenwood command until a Yaesu command has been received or power has been turned OFF then ON)
YB	YB;	YBx;	YB000000x; x = 0-1	0 - Normal dial 1 - Accel Dial
YC	YC;	YCx;	YC000000x; x = 0-1	Auto saving 0 - OFF, 1 - ON
YD	YD;	YDx;	YD000000x; x = 0-1	Fine Tune 0 - OFF, 1 - ON
YE	YE;	YEx;	YE000000x; x = 0-1	Offset type 0 - Single (global) offset, 1 - Per-band (BPF bands) offsets
YF	-	YFxxxxx;	YF000xxxx; (Neg) YF001xxxx; (Pos) xxxx = 0000-9999	Offset quick set (0001 - 9999 kc) Offsets below 0 (Negative) YF0xxxx; Offsets above 0 (Positive) YF1xxxx;
YG	YG;	YGx;	YG000000x; x = 1-4	Function YG1; - VFOA, YG2; - VFOB, YG3; - MR, YG4; - CB, YG5; - CB Toggle
YH	YH;	YHxx;	YH000000xx; xx = 01-40	CB Channel eg YH0000001; YH0000040;
YI	-	YIx;	-	CB Band (looping) 0 - Decrease 1 - Increase
YJ	YJ;	YJx;	YJ000000x;	CB Band (1 - 8) Default: 4 (MID band)
YK	YK;	YKx;	YK000000x; x = 1-8 (baud index)	Baud rate YK1; - 1200, YK2; - 2400, YK3; - 4800, YK4; - 9600, YK5; - 19200, YK6; - 38400, YK7; - 57600, YK8; - 115200
YL	YL;	YLx;	YL000000x;	0 - VFD display OFF 1 - VFD display ON (Default)
YM	YM;	Ymx;	YM000000x; x = 0,1	0 - Virtual PMS switch OFF 1 - Virtual PMS switch - ON 2 - Virtual PMS switch toggle OFF/ON
YN	YN;	YNx;	YN000000x; x = 0,1	0 - Virtual 500k switch ON 1 - Virtual 500k switch OFF 2 - Virtual 500k switch toggle OF/ON
YO	YO;	YOx;	Y0000000x x = 0,1,2,3	0 - Offsets disabled, 1 - Offsets applied to TX ONLY, 2 - Offsets applied to RX only, 3 - Offsets applied to TX and RX (default)
YP	-	YP;	-	SPLIT button
YQ	-	YQ;	-	MR/VFO button
YR	-	YR;	-	VFO>M button
YS	-	YS;	-	D LOCK button
YT	-	YT;	-	VFO A/B button
YU	-	YU;	-	Hidden Wideband / Gen TX switch (default after reboot)
YV	-	YV;	-	M>VFO button
YW	-	YW;	-	BAND / CHANNEL UP button (excludes CB band - See YI / YJ)
YX	-	YX;	-	BAND / CHANNEL DOWN button (excluding CB band)
YY	-	YY;	-	CLAR button
YZ	-	YZx;	-	0 - VFO<>M button, 1 - VFOA<>VFOB (cat only)

Errata

v1.3 PCB FTPlus boards have serial RX and TX now in Microchip PIC naming order, so are thus flipped

Version numbering:

New versions of FTPlus are given a new version number, eg 1.4... 1.4.5 etc

Bugfix versions keep the same version number and have a letter, eg 1.4a, 1.4b, 1.4.5a etc etc...

The supplement version will always match the VERSION NUMBER. A new supplement will be released at the same time as the new firmware VERSION NUMBER. For bugfix releases, the supplement is not updated.

Changelogs

v1.7.0 – Mostly CAT changes

- Cat TX in CLAR or SPLIT modes now works correctly
- Cat driver improvements – Commander in Yaesu mode works correctly now
- Y-Library introduction:
 - Complete front panel control with PMS and 500k ‘virtual’ buttons
 - Complete FTPlus command control over CAT
 - Statuses returned on required functions
 - Offset / RIT information is now controllable over CAT

Credits:

Daniel Keogh

PLL information

Prototyping

Siegmund Souza

Marco Steneker

Testing and research

Matthew Bostock

Programming

Board design

Documentation

Thanks for all the encouragement during developing this.

TRAV Industries 2026